# Lab M8.02 – Data Warehouse Fundamentals & Star Schema Design

**Step 1: Fact vs Dimension Identification**

Order amount

- Fact or Dimension? Fact
- Reasoning: Why? Measurable
- How is it used in analytics? aggregations

Quantity sold

- Fact or Dimension? Fact
- Reasoning: Why? Measurable
- How is it used in analytics? aggregations

Customer

- Fact or Dimension? Dimensions
- Reasoning: Why? Descriptive attribute
- How is it used in analytics? Filtering, GROUP BY

Product

- Fact or Dimension? Dimensions
- Reasoning: Why? Descriptive attribute
- How is it used in analytics? Filtering, GROUP BY

Date

- Fact or Dimension? Dimensions
- Reasoning: Why? Descriptive attribute
- How is it used in analytics? Filtering, GROUP BY, trends

Region

- Fact or Dimension? Dimensions
- Reasoning: Why? Descriptive attribute
- How is it used in analytics? Filtering, GROUP BY

**Step 2: Fact Table Design**

fact_sales

- sale_id (PK, surrogate key)
- customer_id (FK - dim_customer)
- product_id (FK - dim_product)
- date_id (FK - dim_date)
- region_id (FK - dim_region)
- order_amount (measure)
- quantity (measure)

- discount (measure)
- profit (measure)

Foreign keys are needed to connect with dimension tables. One row represents one sale. The grain is appropriate to allow aggregation by any dimension.

**Step 3: Dimension Tables Design**

dim_customer

- customer_id (PK, surrogate)
- customer_name
- email
- segment
- registration_date
- city
- state

dim_product

- product_id (PK, surrogate)
- product_name
- category
- subcategory
- brand
- unit_price
- unit_cost

dim_date

- date_id (PK, surrogate)
- date
- year
- quarter
- month
- week
- day_of_week
- is_holiday

dim_region

- region_id (PK, surrogate)
- region_name
- country
- state
- city
- timezone

Step 4: Star Schema Diagram

```
                          dim_customer
                          -------------
                          PK customer_id
                          customer_name
                          email
                          segment
                          registration_date
                          city
                          state
                          country
                                 |
                                 |
                                 |
dim_product  -----------   fact_sales   -----------  dim_date
-------------              -----------               ----------
PK product_id              PK sale_id               PK date_id
product_name               FK customer_id           date
category                   FK product_id            year
subcategory                FK date_id               quarter
brand                      FK region_id             month
unit_price                 order_amount             week
unit_cost                  quantity                 day_of_week
                           discount_amount          is_holiday
                           profit_amount            is_weekend
                                 |
                                 |
                                 |
                          dim_region
                          -------------
                          PK region_id
                          region_name
                          country
                          state
                          city
                          timezone
```